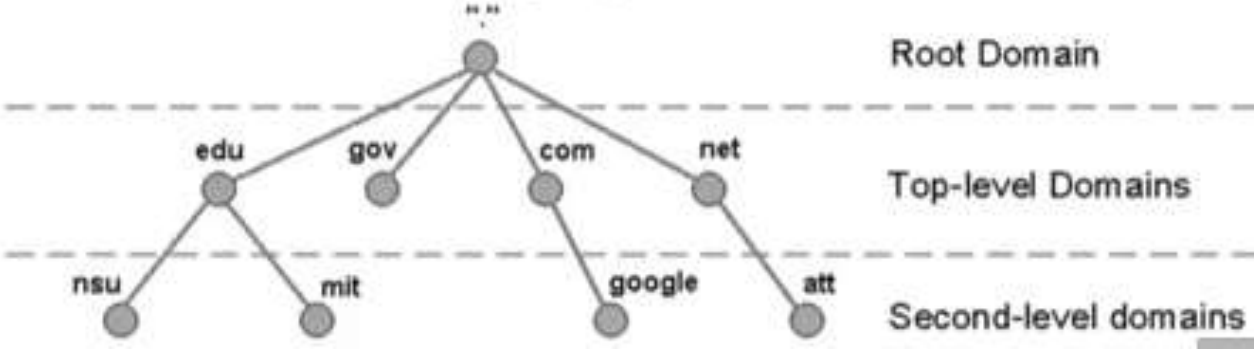
Domain Name System (DNS)

- What is DNS?
  - DNS is primarily used for host name to IP resolution.
  - Not centralized. Highly distributed.



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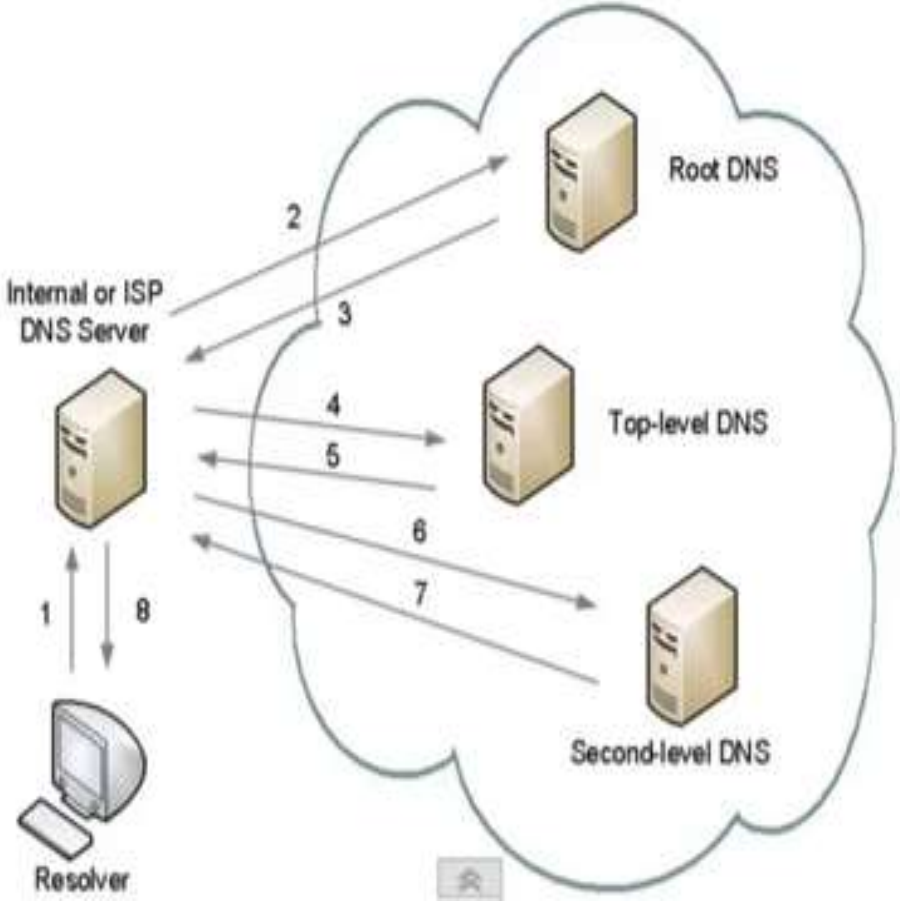


Subscri

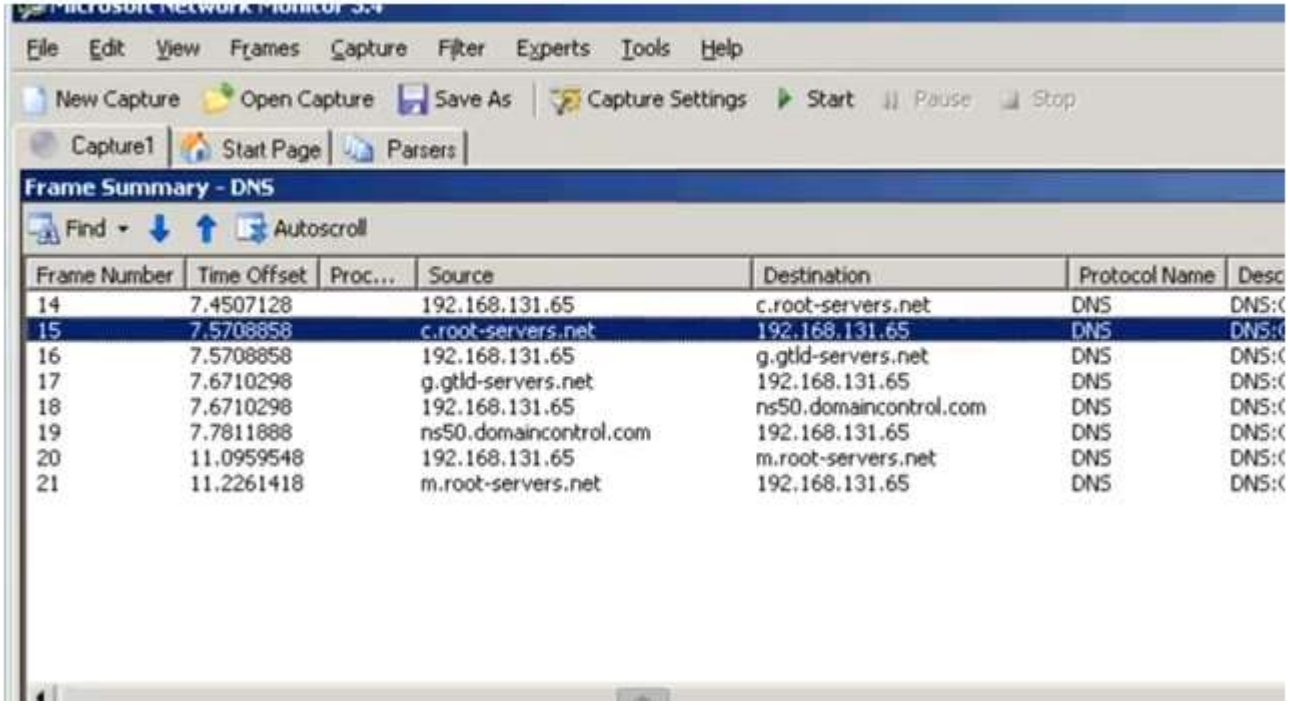
Command Prompt - nslookup

```
C:\>nslookup
Default Server:  ns1.itgeared.test
Address:  192.168.131.65

> www.itgeared.com
Server:  ns1.itgeared.test
Address:  192.168.131.65

Name:    itgeared.com
Address:  184.168.250.1
Aliases:  www.itgeared.com

>
```



Frame Number	Time Offset	Proc...	Source	Destination	Protocol Name	Desc
14	7.4507128		192.168.131.65	c.root-servers.net	DNS	DNS:(
15	7.5708858		c.root-servers.net	192.168.131.65	DNS	DNS:(
16	7.5708858		192.168.131.65	g.gtld-servers.net	DNS	DNS:(
17	7.6710298		g.gtld-servers.net	192.168.131.65	DNS	DNS:(
18	7.6710298		192.168.131.65	ns50.domaincontrol.com	DNS	DNS:(
19	7.7811888		ns50.domaincontrol.com	192.168.131.65	DNS	DNS:(
20	11.0959548		192.168.131.65	m.root-servers.net	DNS	DNS:(
21	11.2261418		m.root-servers.net	192.168.131.65	DNS	DNS:(

Definition: *nslookup* is a network utility program used to obtain information about Internet servers. As its name suggests, the utility finds name server information for domains by querying DNS.

Using nslookup

Most computer operating systems include a built-in command line program called nslookup.exe. Some network providers also host their own Web-based services of this same utility. These programs are all designed to perform name server look-ups against specified Internet domains.

To use the Windows version, open a command prompt and launch the tool as follows:

```
C:\> nslookup
Default Server: resolver1.opendns.com
Address: 208.67.222.222

\>
```

This first step identifies which DNS server configured on the PC is in use for look-ups.

IP Address Lookup

By default, nslookup responds with the primary IP address associated with a domain specified. To query the primary address of about.com, for example:

```
\> about.com
Address: 67.215.65.132
```